

Rationality in Practice: CCEI & Revealed Preference

Discussion Review and Problem Set Analysis

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Discussion Board Highlight: The Malawi Study

Student Question

“The CCEI improvement was driven by 9th graders who were already close to consistent decision-making, not the least rational ones. Are we really seeing 'rationality' improve, or are we just helping students who already almost understood it? And how much can we trust CCEI to generalize to contexts like financial literacy?”

- **Why this is an excellent critique:** It directly challenges the *external validity* of our metric and strikes at the core debate of how empirical economics defines rationality.

1. Rationality: A Degree, Not a Binary

- **Standard Neoclassical Theory:** Rationality (GARP) is a strict Yes/No binary. Make one mistake, and you are deemed “irrational.”
- **Empirical Economics (CCEI):** The Critical Cost Efficiency Index transforms rationality from a brittle binary into a continuous metric—a *degree* of rationality.
- **The Treatment Effect:** Because the tuition support intervention mostly helped those already close to the threshold, it confirms that interventions do not rewrite a person’s foundational logic from “irrational” to “rational.” Instead, they incrementally shift the *degree* of efficiency.

2. Relieving Cognitive Load vs. “Teaching” a Task

- **The Intervention Mechanism:** The Malawi intervention was tuition support. It did not explicitly “teach” rationality or decision-making.
- **Scarcity Mindset:** Poverty and financial stress impose a heavy, continuous cognitive tax on the brain’s bandwidth.
- **Clearing the RAM:** The intervention likely relieved just enough financial stress for students to clearly apply the reasoning skills they already possessed. It didn’t install new cognitive software; it cleared the RAM so their existing hardware could run efficiently.

3. Does CCEI Generalize? (Rationality vs. Literacy)

- **External Validity:** CCEI is a strong predictor of total household wealth in large longitudinal datasets, independent of education or math ability. Individuals with low CCEI scores are prone to costly “money pump” behaviors in the real world (e.g., carrying credit card debt while holding cash in a checking account).
- **The Distinction:** *CCEI (Rationality)* measures whether your choices are free from logical contradictions. *Financial Literacy* measures domain-specific knowledge (like understanding compound interest).
- **The Takeaway:** A perfect CCEI won't save you if you don't know the rules of finance. But improving baseline consistency ensures that when you *do* learn financial concepts, you won't contradict yourself and leave money on the table.

Practice Problem

Jennie's Consumption Data

Jennie's consumption behavior is summarized by the following three observations, where $x^t = (x_1^t, x_2^t)$ denotes the consumption bundle and $p^t = (p_1^t, p_2^t)$ denotes the corresponding price vector in period t :

$$x^1 = (4, 2), \quad p^1 = (2, 1)$$

$$x^2 = (2, 2), \quad p^2 = (1, 2)$$

$$x^3 = (1, 2), \quad p^3 = (3, 1)$$

- ❶ (10 points) Do Jennie's choices violate the Weak Axiom of Revealed Preference (WARP)? Fully justify your answer.
- ❷ (10 points) Do Jennie's choices violate the Generalized Axiom of Revealed Preference (GARP)? Fully justify your answer.
- ❸ (10 points) Calculate Jennie's Critical Cost Efficiency Index (CCEI). What does the CCEI imply about Jennie's level of economic rationality?